BIOCHEMISTRY CUMULATIVE EXAM

March 2, 1996

A. (20 points) Many membrane proteins are thought to be bound into biological membranes via transmembrane helical segments of 19-21 amino acids in length. Why are most transmembrane helices of the length indicated? When transmembrane helices of differing lengths are found, they are generally longer but seldom shorter than this length. Suggest a reason why this is observed.

B. (20 points) Although the structures of only a few membrane proteins are known, predictions are often made based on hydropathy plots determined on the basis of protein or gene sequences as to the location of the transmembrane regions in the protein structure. Describe the plotting method used to make such predictions.

C. (40 points) Consider a protein which has two transmembrane helices. Draw the possible ways it might be oriented in the membrane. Describe in detail at least two experimental methods which might be used to determine which orientation is correct. Keep in mind that the membrane in which the protein is located probably has hundreds of different proteins associated with it in addition to the one in which you are interested.

D. (20 points) The genes for many membrane proteins often encode a precursor form of the protein which is longer than the final, mature form of the protein - usually with an extension at the N-terminal end of the protein. What might be the purpose of this N-terminal extension? How might you test whether your proposed purpose is correct?